ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PRACTICE EXAM 3

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 8 year: designing a footbridge, struts, ties and trusses, plastics and alloys, a bicycle gear and speed problem, belt and gear drives, pulley systems, an automatic watering system with AND logic, sensors, and energy-saving lighting.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings and tables.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which of these is a shell structure?

A Pylon B Car body C Dam wall D Ladder

1.2 Triangles are used in truss bridges mainly to make them …

A heavier. B flexible. C colourful. D rigid.

1.3 When two gears of different sizes mesh, the bigger gear turns …

A slower than the smaller gear. B faster than the smaller gear.
C at the same speed. D in the same direction.

1.4 Which mechanism is used to drive the back wheel of a bicycle?

A Belt drive B Rack and pinion C Chain drive D Worm gear

1.5 A straight bar with teeth cut along it is called a …

A pinion. B sprocket. C rack. D cam.

1.6 A machine may only start when both safety switches are closed. This is …

A AND logic. B OR logic. C NOT logic. D no logic.

1.7 The SI unit of electrical resistance is the …

A volt. B ampere. C watt. D ohm.

1.8 Polystyrene is an example of a …

A thermosetting plastic. B thermoplastic.
C metal. D composite.

1.9 The view of an object drawn as seen from directly in front is the …

A plan. B side view. C isometric view. D front view.

1.10 In an isometric drawing, the horizontal edges are drawn at …

A 15° B 30° C 45° D 90°

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QUESTION 2: STRUCTURES

2.1 A rural community needs a footbridge across a stream that is 12 m wide. Learners and older people will use it every day.

2.1.1 Suggest a suitable type of bridge and give a reason for your choice. (2)

2.1.2 Suggest TWO suitable materials for the bridge and give a reason for each. (2)

2.1.3 Name TWO forces that the bridge must be able to withstand. (2)

2.1.4 Give TWO ways to make the bridge more stable in strong wind and floods. (2)

2.1.5 Give TWO safety features the bridge should have. (2)

2.2 Explain the difference between a strut and a tie. (2)

2.3 Draw a neat, labelled sketch of a simple truss bridge. Show the deck, the supports and the triangles. (4)

2.4 What can happen to a structure if the forces on it are too large for its materials? (2)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 Give ONE everyday use of each plastic: (a) PET; (b) HDPE; (c) PVC; (d) polystyrene. (4)

3.2 Why are thermosetting plastics used to make electrical plugs and pot handles? (2)

3.3 South Africa charges a levy (small tax) on plastic shopping bags. What is the main purpose of this levy? (1)

3.4 Give TWO ways in which learners can reduce plastic waste at school. (2)

3.5 What is a bioplastic? (1)

3.6 What is an alloy? Give ONE example. (2)

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QUESTION 4: MECHANICAL SYSTEMS

4.1 A bicycle has a front chainring with 48 teeth and a rear sprocket with 16 teeth.

4.1.1 How many times does the rear wheel turn for each full turn of the pedals? (2)

4.1.2 The cyclist pedals at 60 r/min. Calculate the speed of the rear wheel in r/min. (2)

4.1.3 The circumference of the wheel is 2 m. Calculate how far the bicycle travels in one minute and its speed in km/h. (3)

4.1.4 Why does a cyclist use a "high" gear like this on a flat road? (2)

4.2 Explain the difference between the driver and the driven gear in a gear system. (2)

4.3 A pulley system with three supporting ropes lifts a load of 600 N. Ignoring friction, calculate the effort needed. (2)

4.4 Give ONE advantage of a belt drive and ONE advantage of a gear drive. (2)

4.5 Draw two pulleys connected by (a) an open belt and (b) a crossed belt. Show with arrows the direction in which each pulley turns. (2)

4.6 What is torque? (1)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 A water pump in the school garden must switch on only when a timer switch is ON and a soil sensor shows that the soil is dry.

5.1.1 Which type of logic is needed? (1)

5.1.2 Draw and complete a truth table for the system. (4)

5.1.3 Draw a circuit diagram with a battery, the two switches and the pump motor. (4)

5.2 Name TWO sensors (input devices) that can be used in electronic circuits, and what each one detects. (2)

5.3 What happens to the brightness of bulbs when more identical bulbs are added (a) in series and (b) in parallel to the same battery? (2)

5.4 An LED bulb uses 9 W and an old filament bulb uses 60 W to give the same light. Each bulb is switched on for 5 hours. Calculate how much energy (in kWh) is saved by using the LED. Use: energy = power × time. (3)

5.5 Which component is shown by the symbol of a circle with the letter M inside it? (1)

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TOTAL: 75

END OF QUESTION PAPER